Institute Vision, Mission, Quality Policy, Core Values

Vision of the Institute

Acharya Institute of Technology, committed to the cause of sustainable value-based education in all disciplines, envisions itself as a global fountainhead of innovative human enterprise, with inspirational initiatives for Academic Excellence.

Mission of the Institute

Acharya Institute of Technology strives to provide excellent academic ambience to the students for achieving global standards of technical education, foster intellectual and personal development, meaningful research, ethical, and sustainable service to societal needs.

Quality Policy

We at Acharya Institute of Technology, promise to continually strive towards total quality in all our endeavours through Equity with Accessibility, Commitment with Honesty, Adaptability with Efficiency while blending Concern for Environment and Social Development.

Core Values

1. Pursuit of Excellence
2. Integrity and Transparency
3. Leadership

Vision of the Department

To establish as a centre of excellence in moulding young professionals in the domain of Artificial Intelligence and Machine Learning who are capable of contributing to the wellness of the society as well as the industry.

Mission of the Department

1. To nurture the students with multi-disciplinary skills to be able to build sustainable solutions to engineering problems which in turn helps them to grab dynamic and promising career in the industry.
2. To mould energetic and committed engineers by inculcating moral values of professional ethics, social concerns, environment protection, sustainable solutions and life-long learning.

To facilitate research cooperation in the fields of artificial intelligence, machine learning, and data analytics by handholding with national and international research organizations, thus enabling the students to analyse and apply various levels of learning.

Program Educational Objectives (PEOs)

Engineering Graduates will

1. Have a strong foundation in key concepts and techniques related to artificial intelligence and its related fields, thus enabling them to analyze complex problems in various domains and apply AI and ML techniques to develop innovative and effective solutions.
2. Be conscious of the ethical implications of AI/ML technologies, be aware of potential biases and ensure fairness, transparency, and accountability and demonstrate a commitment to develop AI/ML applications for the betterment of society.
3. Possess effective communication skills and be capable of working collaboratively in multidisciplinary teams to address real-world challenges.
4. Have an entrepreneurial mindset and be equipped with the skills and knowledge to create innovative AI/ML solutions and potentially start their own ventures.
5. Demonstrate a commitment to lifelong learning and stay updated with advancements in AI and ML technologies and industry trends to emerge as leaders

in their fields of expertise and domain that support service and economic development of the society.

Program Outcomes (POs)

Engineering Graduates will be able to:

1. **Engineering Knowledge:** Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
2. **Problem Analysis:** Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
3. **Design / Development of Solutions:** Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
4. **Conduct Investigations of Complex Problems:** Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
5. **Modern Tool Usage:** Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modelling to complex engineering activities with an understanding of the limitations.
6. **The Engineer and Society:** Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
7. **Environment and Sustainability:** Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
8. **Ethics:** Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
9. **Individual and Team-work:** Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
10. **Communication:** Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to

comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

11. **Project Management and Finance:** Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
12. **Life-long Learning:** Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

Program Specific Outcomes (PSOs)

Engineering Graduates will be able to:

1. Apply the Artificial Intelligence and Data Analytics skills in the areas of Disaster Management, Health Care, Education, Agriculture, Transportation, Environment, Society and in other multi-disciplinary areas.
2. Analyze and demonstrate the knowledge of Human cognition, Artificial Intelligence, Machine Learning and Data engineering in terms of real-world problems to meet the challenges of the future.
3. Develop computational knowledge and project management skills using innovative tools and techniques to solve problems in the areas related to Deep Learning, Machine learning, Artificial Intelligence which will lead to lifelong learning.

Program Specific Outcome (PSO) to Mission Mapping			
PSOs \ Mission Statements	**Mission 1: Multi-disciplinary skills for sustainable solutions**	**Mission 2: Ethics, social concerns, sustainability, lifelong learning**	**Mission 3: Research cooperation in AI/ML/Data Analytics**
PSO 1: Apply AI and Data Analytics in various domains	3	2	3
PSO 2: Analyze and demonstrate AI/ML knowledge for real-world challenges	3	2	3
PSO3: Develop computational and project management skills for AI/ML	2	3	3